

Commerce XI Mathematics & Statistics · Part 2

Unit 3: Skewness

A quick reference based on Unit 3, textbook pages 37-44.

How to use this guide

Use	What to check
Before solving	Identify whether the information contains mean, median, mode, S.D., variance or quartiles, then select the appropriate coefficient.
During solving	Use Karl Pearsonian coefficient for mean-mode or mean-median data and Bowley coefficient for quartiles; retain the sign.
Before answering	Round only at the requested final stage and state whether the distribution is positively skewed, negatively skewed or symmetric.

Scope

Meaning and direction of skewness · symmetric and asymmetric distributions · Karl Pearsonian coefficient · empirical relation · Bowley coefficient · grouped distributions · open-ended classes

Textbook Answer Notes

Exercise 3.1, Q3: the printed answer key gives 1. From $Q_1 + Q_3 = 100$, $Q_3 - Q_1 = 40$ and $Q_2 = 50$, Bowley's coefficient is $(100 - 100)/40 = 0$. The App uses the mathematically correct answer 0.
All other verified Exercise 3.1 and Miscellaneous Exercise 3 answers match the textbook answer key.

1. Meaning of skewness

Two distributions can have similar measures of central tendency and dispersion but still differ in symmetry. Skewness measures whether a distribution is symmetric or asymmetric. If it is asymmetric, skewness also describes its direction and extent.

Type	Frequency pattern and graph	Central-value relation	Textbook example
Symmetric	Values equidistant from the mean have equal frequencies; both tails balance.	Mean = Median = Mode	Lengths of rods and weights of individuals are often approximately symmetric.
Positive skew	Lower values carry relatively high frequencies; the right tail is longer and points to higher x-values.	Mean > Median > Mode	Employee salaries: a small number of very high salaries extend the right tail.
Negative skew	Higher values carry relatively high frequencies; the left tail is longer and points to lower x-values.	Mode > Median > Mean	Lifespans: most values may be high, with a smaller number of low values extending the left tail.

Textbook zero-skewness illustration

x	14	16	18	20	22	24	26
f	2	5	10	14	10	5	2

The mean is 20. Values equally distant from 20 have equal frequencies: 18 and 22 each have frequency 10; 16 and 24 each have frequency 5; 14 and 26 each have frequency 2. Therefore the distribution is symmetric.

Textbook positive-skewness illustration

x	14	16	18	20	22	24	26
f	2	8	17	10	7	4	2

2. Karl Pearsonian coefficient of skewness

The signed difference Mean - Mode indicates direction, but it cannot compare distributions measured in different units. Karl Pearson standardized this difference by dividing it by the standard deviation.

Situation	Formula / result
Primary formula	$S_{kp} = (\text{Mean} - \text{Mode}) / \text{S.D.}$
Mode is indeterminate	$\text{Mean} - \text{Mode} = 3(\text{Mean} - \text{Median})$
Alternative formula	$S_{kp} = 3(\text{Mean} - \text{Median}) / \text{S.D.}$
Symmetric	$S_{kp} = 0$
Positive skew	$S_{kp} > 0$
Negative skew	$S_{kp} < 0$
Usual range	Most values lie between -1 and 1
Full range	All values lie between -3 and 3

S_{kp} is independent of the units and size of the data values. Therefore coefficients from different distributions can be compared.

Choosing a Pearson formula

Given	Action
Mean, mode and S.D.	Use $(\text{Mean} - \text{Mode})/\text{S.D.}$ directly.
Mean, median and S.D.	Use $3(\text{Mean} - \text{Median})/\text{S.D.}$
Variance instead of S.D.	First calculate S.D. = positive square root of variance.
Grouped frequency distribution	Calculate mean, mode and S.D.; then apply the primary formula.

Grouped mode formula

Mode = $L + [(f_m - f_1) / (2f_m - f_1 - f_2)] \times h$, where L is the lower limit/boundary of the modal class, f_m its frequency, f_1 the pre-modal frequency, f_2 the post-modal frequency and h the class width.

3. Pearsonian textbook solved examples

Example 1: distribution of profits

Profit (Rs. lakh)	10-20	20-30	30-40	40-50	50-60
No. of firms	12	18	25	10	7

Step	Calculation
Modal class	30-40 because its frequency 25 is the greatest.
Mode	$30 + [(25 - 18)/(2 \times 25 - 18 - 10)] \times 10 = 33.18$
Coding	Take $A = 35$, $h = 10$ and $u = (x - 35)/10$.
Mean	$u\text{-bar} = -18/72 = -0.25$; Mean = $35 + 10(-0.25) = 32.5$
Variance and S.D.	Variance(u) = $104/72 - (-0.25)^2 = 1.3819$; Variance(x) = $100(1.3819) = 138.19$; S.D. = 11.7554
Coefficient	$S_{kp} = (32.5 - 33.18)/11.7554 = -0.0578$
Interpretation	$S_{kp} < 0$, so the distribution is negatively skewed.

Examples 3-5: direct Pearson calculations

Data	Method and result
Mean = 100, Median = 98.5, S.D. = 9	Mode = $100 - 3(100 - 98.5) = 95.5$; $S_{kp} = (100 - 95.5)/9 = 0.5$; positively skewed.
Mode exceeds mean by 7; variance = 100	Mean - Mode = -7 ; S.D. = 10 ; $S_{kp} = -7/10 = -0.7$; negatively skewed.
Mean = 214, Mode = 218, variance = 196	S.D. = 14 ; $S_{kp} = (214 - 218)/14 = -0.2857$; negatively skewed.

4. Bowley coefficient of skewness

Bowley developed a relative measure based on quartiles. It is especially useful when quartiles are available and can also be used when the extreme classes are open-ended, provided the quartile classes can be located.

Item	Formula or interpretation
Formula	$S_{kb} = (Q_3 + Q_1 - 2Q_2) / (Q_3 - Q_1)$
Equivalent numerator	$(Q_3 - Q_2) - (Q_2 - Q_1)$
Positive skew	$Q_3 - Q_2 > Q_2 - Q_1$; hence $S_{kb} > 0$
Symmetric	$Q_3 - Q_2 = Q_2 - Q_1$; hence $S_{kb} = 0$
Negative skew	$Q_3 - Q_2 < Q_2 - Q_1$; hence $S_{kb} < 0$
Range	$-1 < S_{kb} < 1$

Locating quartiles in grouped data

Quartile	Position	Grouped-data formula
Q_1	$N/4$	$L + h[(N/4 - c.f.)/f]$
Q_2	$N/2$	$L + h[(N/2 - c.f.)/f]$
Q_3	$3N/4$	$L + h[(3N/4 - c.f.)/f]$

For each quartile, L is the lower limit/boundary of its quartile class, h is class width, f is the frequency of that class and c.f. is the cumulative frequency before that class.

Symmetric quartile relations

Relation	Meaning
$Q_3 - Q_2 = Q_2 - Q_1$	Q_2 is equally distant from Q_1 and Q_3 .
$2Q_2 = Q_1 + Q_3$	Equivalent symmetric relation.
Q_1, Q_2, Q_3 are in A.P.	The distribution is symmetric.

5. Bowley textbook solved examples

Example 1: $Q_1 = 80$, $Q_2 = 100$, $Q_3 = 120$

$S_{kb} = (120 + 80 - 2 \times 100) / (120 - 80) = 0/40 = 0$. Therefore the distribution is symmetric.

Example 2: open-ended weekly-wage distribution

Wages	Below 300	300-400	400-500	500-600	600-700	Above 700
Workers	5	8	18	35	27	7
Less-than c.f.	5	13	31	66	93	100

Quartile	Class and calculation	Value
Q_1	25th observation; class 400-500; $400 + 100(25 - 13)/18$	466.67
Q_2	50th observation; class 500-600; $500 + 100(50 - 31)/35$	554.28
Q_3	75th observation; class 600-700; $600 + 100(75 - 66)/27$	633.33

$S_{kb} = (633.33 + 466.67 - 2 \times 554.28) / (633.33 - 466.67) = -0.05$ approximately. The distribution is negatively skewed.

Examples 6 and 7

Data	Calculation and result
$Q_1 = 35$, $Q_2 = 40$; distribution symmetric	$Q_3 = 2Q_2 - Q_1 = 80 - 35 = 45$.
$Q_1 = 15$, $Q_2 = 21$, $Q_3 = 29$	$S_{kb} = (29 + 15 - 42) / (29 - 15) = 2/14 = 0.143$; positively skewed.

6. Additional solved examples and reasoning

Question	Reasoning and conclusion
Relation between mean and mode for negative skewness	S.D. is positive and $S_{kp} < 0$. Therefore Mean - Mode < 0 , so Mean $<$ Mode.
Relation among quartiles for symmetry	$S_{kb} = 0$ gives $Q_3 - Q_2 = Q_2 - Q_1$, so $2Q_2 = Q_1 + Q_3$.
Q_1, Q_2, Q_3 are in A.P.	Their consecutive differences are equal, so $S_{kb} = 0$ and the distribution is symmetric.
Mean $>$ Mode	S.D. is positive and Mean - Mode > 0 , so $S_{kp} > 0$ and the distribution is positively skewed.

7. Exercise 3.1 answer map

Question	Verified answer / conclusion
Q1	$S_{kp} = -0.45$
Q2	Mode = 63; Median = 61
Q3	$S_{kb} = 0$ (printed key 1 is inconsistent)
Q4	$Q_1 = 29$
Q5	Coefficient = -0.1881; negatively skewed
Q6	$S_{kp} = 0$
Q7	$S_{kb} = -0.5$
Q8	$S_{kb} = -0.143$ approximately

8. Miscellaneous Exercise 3 answer map

Question	Verified answer
Q1	1
Q2	-1.5
Q3	0.2
Q4	-0.2
Q5	$Q_3 = 70; Q_1 = 30$
Q6	Mode = 195.2; Median = 198.4
Q7	-0.4760
Q8	-0.3194
Q9	-1.3182
Q10	-0.6

9. Activities 3.1-3.2

Activity	Required work
Activity 3.1	Collect Mathematics marks of 12 students, arrange them in ascending order, determine Q_1 , Q_2 and Q_3 , calculate S_{kb} and write a comment from its sign.
Activity 3.2	Plot the supplied (x, f) points, draw a smooth freehand frequency curve, identify the longer tail, state the direction of skewness and write the corresponding relation between mean and mode.

10. Complete problem-solving checklist

Check	Action
Identify the supplied values	Mean/mode/S.D. suggests Pearson; quartiles suggest Bowley.
Variance is supplied	Calculate the positive S.D. before substitution.
Mode is missing	Use $3(\text{Mean} - \text{Median})/\text{S.D.}$ when applicable.
Grouped data	Find the required mean, mode, S.D. or quartiles from the frequency table first.
Open-ended extreme classes	Prefer Bowley when quartile classes can be identified.
Numerator sign	A positive, zero or negative numerator determines the direction of skewness.
Interpret the result	State positive skew, negative skew or symmetry after calculating the coefficient.
Rounding	Keep working values sufficiently accurate; round only the final answer when instructed.
Answer format	Include the minus sign when required. A skewness coefficient is not a percentage.

11. Common mistakes to avoid

Mistake	Correction
Using variance directly as S.D.	Take the square root first.
Reversing Mean - Mode	Pearson uses Mean - Mode, in that order.
Using only $Q_3 - Q_1$	Bowley also needs Q_2 in the numerator.
Ignoring the sign	The sign carries the direction of skewness.
Mixing Pearson and Bowley inputs	Choose one method from the available information.
Treating zero as positive or negative	A zero coefficient indicates a symmetric distribution.

12. One-line revision

Right tail longer: positive skew, Mean > Median > Mode.
Left tail longer: negative skew, Mode > Median > Mean.
Balanced: zero skew, Mean = Median = Mode.
Pearson: mean, mode/median and S.D. **Bowley:** Q_1 , Q_2 and Q_3 .